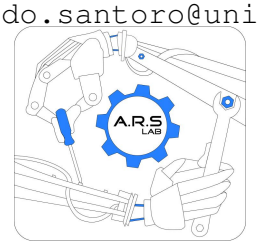
Digital Filters

Corrado Santoro

Dipartimento di Matematica e Informatica
corrado.santoro@unict.it



Advanced Robotics and Autonomous Systems

Notes

Frequency Analysis

Notes

Fourier Analysis

Any periodic signal $s(t)$ with period T (and frequency $f = \frac{1}{T}$) can be decomposed as a **linear combination** of infinite sinusoids and cosinusoids with coefficients a_i and b_i:

$$s(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(\frac{2\pi}{T} nt) + b_n \sin(\frac{2\pi}{T} nt)]$$

Indeed the signal is composed of a weighted sum of components $\cos(\frac{2\pi}{T} t)$, $\cos(\frac{2\pi}{T} 2t)$, $\cos(\frac{2\pi}{T} 3t)$, ... (resp. $\sin$)

Coefficient a_0 is an offset, coefficients $a_1, a_2, \ldots$ and $b_1, b_2, \ldots$ represent the magnitude of each sinusoid/cosinusoid

Notes

Extension to Sample Sequences

Any also non-periodic sequence of samples $s(k)$, with length $k = 1, \ldots, N$, and sampling time ΔT can be extended to periodic by considering infinite repetitions of $s(k)$

To the resulting (periodic) signal $s'(k)$ can be approximated using a fourier series:

$$s'(k) = \frac{a_0}{2} + \sum_{i=1}^{N} [a_n \cos(\frac{2\pi}{N\Delta T}\ i\ k\Delta T) + b_n \sin(\frac{2\pi}{N\Delta T}\ i\ k\Delta T)]$$

Notes

Fourier Analysis and Frequency Spectrum

Spectrum

The **spectrum** of a periodic signal $s(t)$ is a function $S(f)$ that, given a frequency f, returns the magnitude of such a frequency in the signal

Spectrum of Continuous Signal — Fourier Transform

Given a continuous signal $s(t)$, its spectrum is the Fourier Transform:

$$S(f) = \int_{-\infty}^{+\infty} s(t) e^{-2\pi i f t} dt$$

Notes

Fourier Analysis and Frequency Spectrum

Spectrum

The **spectrum** of a periodic signal $s(t)$ is a function $S(f)$ that, given a frequency f, returns the magnitude of such a frequency in the signal

Spectrum of Discrete Signal — Discrete Fourier Transform

Given a discrete signal $s(k)$ of length N its spectrum is the Discrete Fourier Transform:

$$S(k) = \sum_{i=0}^{N-1} s(i) e^{-\frac{2\pi i}{N} n}$$

It is computed using a standard algorithm called **Fast Fourier Transform (FFT)**

Notes

Sampling Time/Frequency

The sampling time/frequency of a signal can be easily computed from the **Sampling/Nyquist Theorem**:

> *Given a signal $s(t)$, whose spectrum is $S(f)$, given that f_{MAX} is the maximum frequency of $s(t)$ that we want to reconstruct, then the sampling frequency must be $f_s = 2f_{MAX}$*
>
> *Given a signal $s(t)$, sampled at frequency f_s, the resulting discrete sequence $s'(k)$ is able to reconstruct $s(t)$ up to frequency $0.5f_s$*
>
> *Frequency $0.5f_s$ is then called the **Nyquist Frequency***

Notes

Filters

Notes

Filters

Filter

A **filter** can be represented as a function of the frequency $H(f)$ able to apply the following transformations:

$$\begin{cases} H(f) & < 1 \quad \text{Frequency } f \text{ is reduced} \\ H(f) & = 1 \quad \text{Frequency } f \text{ is kept as is} \\ H(f) & > 1 \quad \text{Frequency } f \text{ is amplified} \end{cases}$$

Conceptually, given a signal $s(t)$, applying a filter $H(f)$ means to derive the spectrum of the signal $S(f)$ and compute:

$$S'(f) = S(f)\ H(f)$$

The filtered signal $s'(t)$ is obtained by applying the inverse Fourier Transform to $S'(f)$

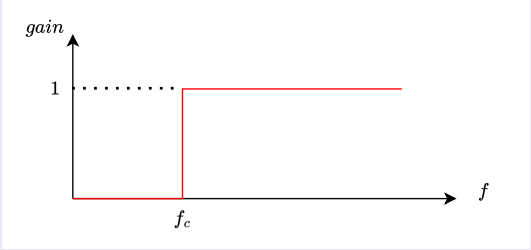
Notes

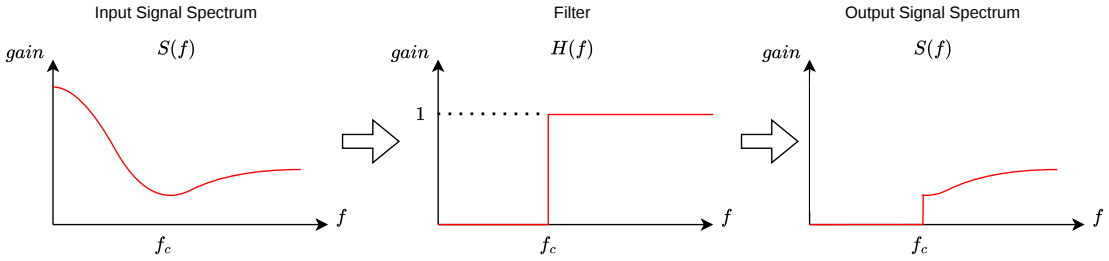
Filters

Filter

Filters are usually represented in *f* / *gain* charts.

Example 1: A filter that cuts all frequencies below f_c and leaves as is all frequencies above f_c



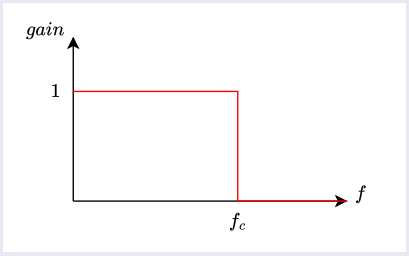


Notes

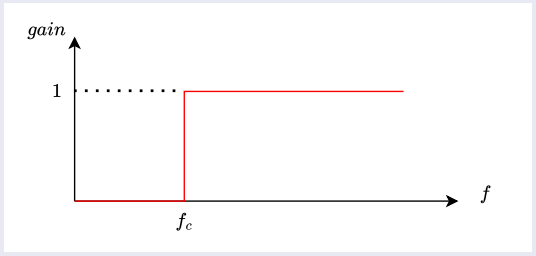
Filters

Basic Ideal Filters

There are four types of basic filters:

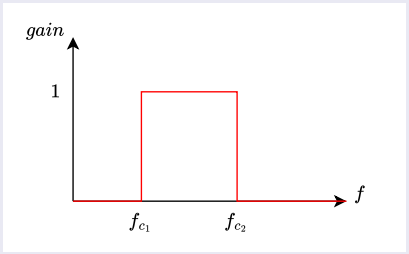


Low Pass

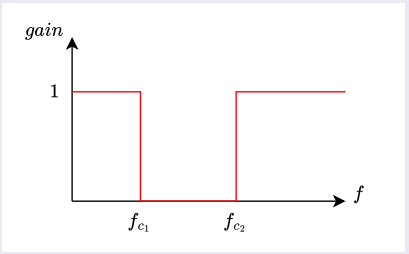


High Pass

f_c is



Band Pass



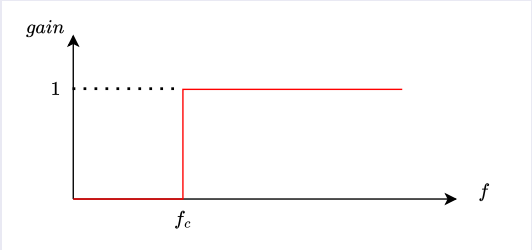
Band Stop

called **cut-off frequency**

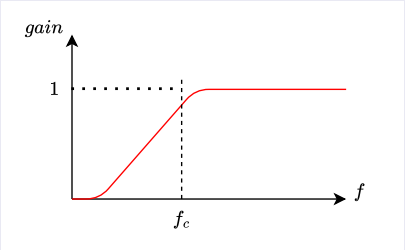
Notes

Real Filters

A real filter never has a sharp transition in the cut-off frequency:



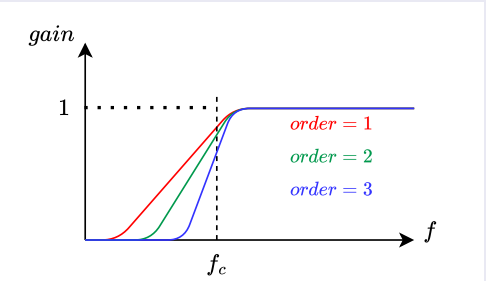
Ideal High Pass



Real High Pass

The transition is gradual and its slope is related to the filter order

The higher the order the better the **selectivity**:



Notes

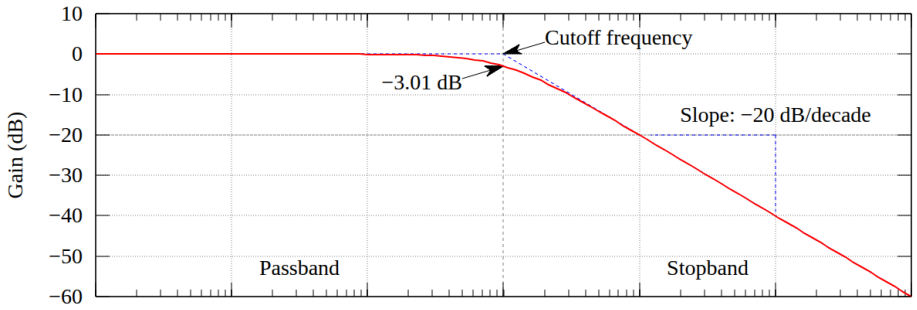
Bode Diagrams

Gain Bode Diagram

- The plot freq/gain is also called **(Gain) Bode Diagram**
- In Bode Diagrams, values in axes are reported in a logaritmic (10-base) scale
- The **gain** is usually reported in **decibel (dB)** using the formula:

$$gain_{dB} = 20log_{10}|H(f)|$$

- cut-off frequency corresponds to a $gain_{dB} = -3dB$, $gain = \frac{1}{\sqrt{2}}$



Notes

Digital Filters and Filter Design

Notes

Digital Filters

A **digital filter** is defined with formula:

$$y(k) + a_1 y(k-1) + a_2 y(k-2) + \cdots + a_n y(k-n) = b_0 u(k) + b_1 u(k-1) + \cdots + b_m u(k-m)$$

Yes, it is an **ARX system model**.

Order is defined by n and m, while the **type** (low-pass, high-pass, etc.) and **cut-off frequencies** are defined in the $a_1, \ldots, b_0, b_1, \ldots$ parameters

Notes

Infinite-Impulse Response

A digital defined as:

$$y(k) + a_1 y(k-1) + a_2 y(k-2) + \cdots + a_n y(k-n) = b_0 u(k) + b_1 u(k-1) + \cdots + b_m u(k-m)$$

is also called **IIR - Infinite Impulse Response**, because its response to impulse **never becomes zero**, i.e. **becomes zero only asymptotically** for $t \to \infty$

Finite-Impulse Response

A digital defined as:

$$y(k) = b_0 u(k) + b_1 u(k-1) + \cdots + b_m u(k-m)$$

is also called **FIR - Finite Impulse Response**, because its response to impulse **sooner or later becomes zero**

Notes

First-Order Low Pass

A **first-order low pass filter** is defined as:

$$y(k) = (1-\alpha)y(k-1) + \alpha u(k)$$

Where α is computed from sample frequency f_s and cut-off frequency f_c as:

$$\alpha = \frac{2\pi\frac{f_c}{f_s}}{1+2\pi\frac{f_c}{f_s}}$$

Notes

First-Order High Pass

A **first-order low pass filter** is defined as:

$$y(k) = \alpha y(k-1) + \alpha u(k) - \alpha u(k-1)$$

Where α is computed from sample frequency f_s and cut-off frequency f_c as:

$$\alpha = \frac{1}{1 + 2\pi \frac{f_c}{f_s}}$$

Notes

First-Order Band Pass

A **first-order band pass filter**, with cut-off frequency f_1 and f_2 ($f_1 < f_2$), can be implemented with:

- a low-pass filter with cut-off f_2, followed by
- a high-pass filter with cut-off f_1.

Notes

First-Order Band Stop

A **first-order band stop filter**, with cut-off frequency f_1 and f_2 ($f_1 < f_2$), can be implemented with:

- a low-pass filter with cut-off f_1, summed with
- a high-pass filter with cut-off f_2.

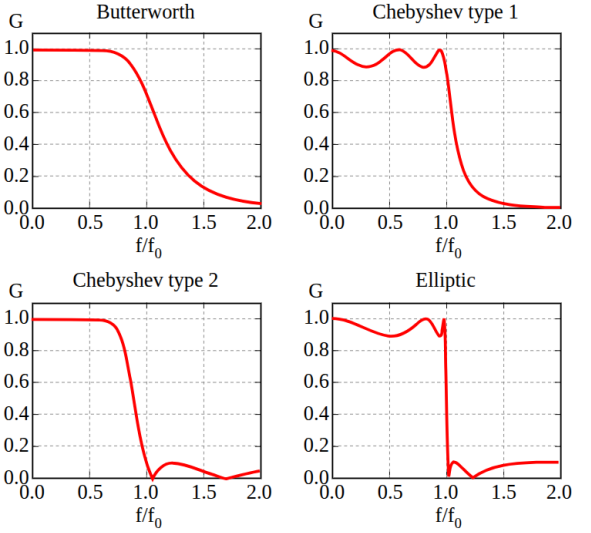
Notes

Special Digital Filters

Notes

Special Digital Filters

- There are several models of n-order filters with $n > 1$
- They are more selective than classical 1-order filters
- Basic models are called Butterworth, Chebyshev and Elliptic (first two from their “inventors”)

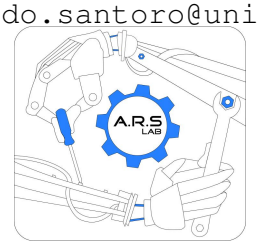


Notes

Digital Filters

Corrado Santoro

Dipartimento di Matematica e Informatica
corrado.santoro@unict.it



Advanced Robotics and Autonomous Systems

Notes